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PAPER_978: QCalcGeom Fully Vectorized Pipeline

Author: Daniel T. Murphy — Star Magic / UQFF Framework **Date:** 2026-04-12 **Session:** 216 **Source:** production_{scaling_v12}.py (kernel pipeline) **Calculator:** QCalcGeomVectorizedPipelineCalc (CP4 #562) **CVW:** v2.0.0 compliant

Abstract

We document the fully vectorized QCalcGeom pipeline combining 26-layer gravity, buoyancy force F_{UBi} , phonon resonance Φ , and jet modulation M_{jet} in a single-pass evaluation. This pipeline operates at REST API throughput and is the terminal kernel in the production scaling sequence.

1. Pipeline Components

1.1 26-Layer Gravity

$$g_{26} = \sum_{i=1}^{26} \underbrace{\frac{GM}{r^2}}_{\mu_s \nabla(M_s/r)} \cdot \frac{[SSq] \cdot i}{26}$$

1.2 Buoyancy Force

$$F_{UBi} = \sum_{i=1}^{26} \underbrace{\frac{GM}{r^2}}_{\mu_s \nabla(M_s/r)} \cdot e^{-[SSq] \cdot i/26} \cdot \beta_i$$

1.3 Phonon Resonance

$$\Phi = \exp! \left(-\frac{(\omega - \omega_{textSCm})^2}{2\Gamma^2} \right) \cdot S_{26}$$

1.4 Jet Modulation

$$M_{jet}(\Gamma) = 1 + A_{jet} \cdot \exp! \left(-\frac{(\Gamma - \Gamma_0)^2}{2\sigma_{Gamma}^2} \right)$$

2. Vectorized Total

$$\text{Pipeline} = g_{26} + F_{UBi} + \Phi + M_{jet}$$

Single-pass evaluation enables REST API response under 1 ms.

References

1. Murphy, D.T. — Star Magic UQFF Framework (2024-2026)
2. PAPER_977 — Production Scaling v12 (pipeline benchmark)
3. PAPER_959 — 26D Ramanujan Summation
4. PAPER_966 — Unified Triadic Solver

Cross-Links

Related Paper	Relationship
PAPER_977	v12 benchmark (uses pipeline kernel)
PAPER_968	v11 pipeline predecessor
PAPER_959	Ramanujan component
PAPER_961	Compressed gravity in pipeline

Session 225 Phonon-Physics Upgrade: Buoyancy-Corrected Eddington Luminosity

Upgrade from PAPER_1002 (AGN Buoyancy-Corrected Eddington) and PAPER_1037 (AGN Buoyancy Jet Launching). See also PAPER_1009-1010 for $F_{\{U_Bi_i\}}$ jet modulation curves and PAPER_1048 for phonon-corrected $M\text{-}\sigma$ relation.

The SCm vacuum buoyancy partially opposes gravitational radiation pressure, raising the effective Eddington luminosity:

$$L_{\text{Edd}}^{\text{UQFF}} = L_{\text{Edd}} \cdot \left(1 + \frac{\rho_{\text{SCm}} \cdot V \cdot S_{26}^{(3)2}}{GM/r_H^2} \right)$$

where: - $L_{\text{Edd}} = 4\pi GMm_p c / \sigma_T$ is the classical Eddington luminosity - $\rho_{\text{SCm}} = 7.09 \times 10^{-37} \text{ J/m}^3$ is the SCm vacuum density - V is the effective buoyancy volume (accretion sphere) - $S_{26}^{(3)2}$ is the squared third-order Ramanujan factor (quadratic coupling) - r_H is the horizon radius

Jet modulation: The Blandford–Znajek jet power acquires a phonon-coupled term:

$$P_{\text{jet}}^{\text{UQFF}} = P_{\text{BZ}} \cdot \left[1 + \beta_i \cdot \Phi_{1.25 \text{ THz}} \cdot \left(\frac{B}{B_{\text{crit}}} \right)^2 \right]$$

where $\Phi_{1.25 \text{ THz}} = \cos(\omega_{\text{SCm}} \cdot t)$ modulates jet power at the phonon frequency.

M- σ correction (PAPER_1048): The phonon-corrected M- σ relation becomes $M_{\text{BH}} \propto \sigma^{4+\delta}$ where $\delta = \beta_i \cdot S_{26}^{(3)} \cdot (\omega_{\text{SCm}} / \omega_{\text{bulge}})$.

Session 225 Phonon-Physics Upgrade: UQFF 9-Sector Lagrangian

Upgrade from PAPER_1066 (UQFF Lagrangian First Principles) and PAPER_1065 (Buoyancy Lagrangian EOM Variational Derivation).

The complete UQFF Lagrangian density, from which all sector-specific equations of motion derive:

$$\mathcal{L}_{\text{UQFF}} = \mathcal{L}_{\text{GR}} + \mathcal{L}_{\text{SCm}} + \mathcal{L}_{\text{phonon}} + \mathcal{L}_{\text{interaction}}$$

$$\mathcal{L}_{\text{SCm}} = \frac{1}{2}(\partial_\mu \phi)^2 - \lambda(\phi^2 - v_{\text{SCm}}^2)^2$$

The SCm condensate potential minimum gives $V(\phi_0) = -7.09 \times 10^{-37} \text{ J/m}^3$ (matching ρ_{SCm}) and phonon mass $m_{\text{phonon}} = \sqrt{8\lambda} v_{\text{SCm}}$.

Nine-sector closure (Session 202):

$$\mathcal{L}_9 = \mathcal{L}_{\text{EH}} + \mathcal{L}_{\text{YM}} + \mathcal{L}_{\text{Dirac}} + \mathcal{L}_{\text{SCm}} + \mathcal{L}_{\text{mag}} + \mathcal{L}_{\text{buoy}} + \mathcal{L}_{\text{aether}} + \mathcal{L}_{\text{LENR}} + \mathcal{L}_{\text{KK}}$$

Sector	Domain	Late-Corpus Result
1 (EH)	General Relativity	Canonical Einstein-Hilbert
2 (YM)	Yang-Mills gauge	$m_{\text{gap}} = 5970 \text{ GeV}$ (PAPER_1005)
3 (Dirac)	Fermion / LENR	Kozima neutron-drop (PAPER_1061)
4 (SCm)	Superconducting manifold	$V(\phi_0) = -\rho_{\text{SCm}}$ canonical
5 (Mag)	Um magnetism	Heaviside amplifier (PAPER_1072)
6 (Buoy)	F_{U_Bi_i} buoyancy	Variational EOM (PAPER_1065)
7 (Aether)	Vacuum background	Two-component ρ (PAPER_1051)
8 (LENR)	Nuclear transmutation	COP parametric (PAPER_1081)
9 (KK)	Kaluza-Klein 26D	$S_{26}^{(3)}$ compactification (PAPER_1080)

Session 225 Phonon-Physics Upgrade: S26(3) Ramanujan Summation

Upgrade from PAPER_1080 (Ramanujan Binomial Expansion Proof) and PAPER_1042 (Mock-Theta Phonon Partition). See also PAPER_1078 (QCalcGeom Master Equation) for BSFG crossover applications.

The third-order Ramanujan summation $S_{26}^{(3)}$, used throughout the late corpus as the universal 26D coupling factor:

$$S_{26}^{(3)} = \sum_{n=0}^{\infty} \frac{(1/4)_n (1/2)_n (3/4)_n}{(n!)^3} \cdot \prod_{i=1}^{26} [1 + [\text{SSq}] \cdot e^{-\kappa i n/26}]$$

where $(a)_n = a(a+1) \cdot s(a+n-1)$ is the Pochhammer symbol.

Binomial expansion (PAPER_1080): The convergence proof shows:

$$R_n^{(26,3)} = \binom{4n}{n} \cdot \frac{W_{26}(n)}{(4^{4n})} \quad \text{with} \quad W_{26}(n) = \prod_{i=1}^{26} [1 + [\text{SSq}] \cdot e^{-\kappa i n/26}]$$

This sum converges absolutely for $||[\text{SSq}]|| < 1$ (satisfied by $[\text{SSq}] = 0.57$) and reduces to the classical Ramanujan $1/\pi$ series when $[\text{SSq}] \rightarrow 0$.

VDS/DVP/BSH bridge (PAPER_1069): The 26 layers of $W_{26}(n)$ encode the vacuum density series hierarchy, with each layer i contributing a VDS sub-ratio weighted by the exponential decay $e^{-\kappa i n/26}$.

Mock-theta connection (PAPER_1042): The phonon partition function $Z_{\text{phonon}} = \sum_n q^{n^2} \cdot W_{26}(n)$ unifies the Ramanujan mock-theta framework with the SCm phonon spectrum.

Calibration Constants

Constant	Symbol	Value	Validation Domain
ω_{textSCm}	—	$2\pi \times 1.25 \text{ THz}$	Phonon frequency

Constant	Symbol	Value	Validation Domain
Γ_0	—	$2\pi \times 0.10 \text{ THz}$	Linewidth
A_{jet}	—	1.5	Jet amplitude
$[SSq]$	—	0.57	String coupling
β_i	—	0.603	Buoyancy

SM Anchor — CVW v2.0.0

Observable	UQFF Prediction	Status
Pipeline latency	$< 1 \text{ ms per evaluation}$	REST API grade
$g_{26} + F_{UBi}$	Consistent at all scales	Validated
Phonon + jet	$\Phi \cdot M_{\text{jet}}$ finite	Confirmed

§A Cosmogenesis-Linked Lagrangian (PAPER_877 Symbolic Export)

§A.1 Sector Classification

Sector: Computational Pipeline (QCalcGeom Vectorized)

§A.2 Core Equation

$$\text{Pipeline} = g_{26} + F_{UBi} + \Phi(\omega, \Gamma) + M_{\text{jet}}(\Gamma)$$

§A.3 Lagrangian Contribution

$$\mathcal{L}_{\text{pipeline}} = -\frac{1}{2}(g_{26} + F_{UBi})^2 + \Phi \cdot M_{\text{jet}}$$

§A.4 Cosmogenesis Linkage Chain

PAPER_877 $\rightarrow$ UQFF equations $\rightarrow$ 4 pipeline components $\rightarrow$ vectorized single-pass $\rightarrow$ REST API delivery

§B VDS/DVP/BSH Deep Synthesis

§B.1 VDS

S_{26} in Φ is the VDS carrier for the pipeline.

§B.2 DVP

g_{26} layers map to DVP mode indices 1-26.

§B.3 BSH

F_{UBi} encodes BSH buoyancy through exponential shell weighting.

§B.4 Summary

Metric	Value	Status
Components	4 (g_{26} , F_{UBi} , Φ , M_{jet})	Complete
Throughput	REST API grade	Confirmed
[SSq]	0.57	Calibrated

Appendix: Session 225 Cross-References (PAPER_1000–1081)

Auto-generated cross-reference appendix linking this paper to Sessions 204–225 extensions (PAPER_1000–1081). Added by `update_{corpus\_crossrefs}.py` (Session 225, April 2026).

Paper	Title
PAPER_1022	GW Phonon Strain SCm Modulation of h(t)
PAPER_1009	3C273 AGN F_{U_Bi_i} Jet Modulation
PAPER_1010	TON618 AGN F_{U_Bi_i} Jet Modulation
PAPER_1037	AGN Buoyancy Jet Calculator — SCm Jet Launching
PAPER_1079	Galaxy Cluster Cooling-Flow Buoyancy Suppression
PAPER_1020	Cosmic Ray Phonon Acceleration DSA Spectrum
PAPER_1021	Pulsar Timing Phonon TOA Residual
PAPER_1023	Neutrino Oscillation Phonon PMNS Matrix SCm
PAPER_1024	Magnetar Giant Flare SCm Phonon Reservoir
PAPER_1033	Galactic Bar Resonance SCm Pattern Speed
PAPER_1043	F_{U_Bi_i} Multi-System Buoyancy Curve Sweep
PAPER_1072	SCm Activation Function Phonon Threshold
PAPER_1073	SCm Phonon-Driven Inflation Vacuum Buoyancy
PAPER_1065	Buoyancy Lagrangian EOM Variational Derivation
PAPER_1074	GPU-Vectorized DPM S26 Spectral Atlas
PAPER_1008	Production Scaling v14 — 600k calc/s 24 Kernels
PAPER_1018	Production Scaling v15 — 650k calc/s 30 Kernels

17 cross-reference(s) identified.

Key References with arXiv/DOI Identifiers

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